Practice questions

Computer Networks 1B

20 questions | 100 marks | 5 marks per question

Answer all questions on separate paper. No calculations are required. Use the mark allocations to guide the depth of your answers. The questions cover all six learning units and are for revision, not an exam prediction.

Question 1 Shared wireless communication

Suppose several devices transmit at the same time over a CDMA wireless link. The receiver knows the code assigned to the device whose data it wants to recover.

a) Explain how the receiver uses that code to recover the required signal. (3)

b) State why suitable codes and control of transmission power are important when several devices share the link. (2)

Question 2 The hidden terminal problem

Suppose two wireless laptops can each communicate with the same access point, but cannot detect one another's transmissions.

a) Explain why checking that the channel appears idle does not prevent all collisions in this situation. (3)

b) Briefly describe how the RTS/CTS mechanism can reduce this problem. (2)

Question 3 Mobility and application continuity

Read the following statement and answer the questions that follow.
A mobile device may remain connected to a network while the quality of an ongoing application changes as the user moves.

a) Explain two reasons why an ongoing voice call may briefly deteriorate during movement between coverage areas. (4)

b) State why maintaining network attachment alone does not guarantee uninterrupted application performance. (1)

Question 4 Interpreting network performance

Suppose a user downloads a large file quickly, but experiences delayed responses in an interactive online game on the same connection.

a) Explain why the download result does not, by itself, establish that the connection is suitable for the game. (2)

b) Identify two other performance measures relevant to the game and briefly describe the effect of each on the user's experience. (3)

Question 5 Content distribution networks

A university stores recorded lectures on a single server. Students in distant regions experience poor playback during busy periods.

a) Describe how placing copies of the lectures at CDN edge servers changes the delivery path to students. (3)

b) Explain one limitation that can remain when a student uses a CDN through a congested local access link. (2)

Question 6 DNS records after a server move

Suppose a website moves to a new server. Its authoritative DNS record is updated, but some users continue to reach the old server for a while.

a) Explain how a cached DNS record and its time to live can account for this behaviour. (3)

b) Briefly explain one preparation the administrator could make before the move to reduce this problem. (2)

Question 7 Cookies and server side sessions

Read the following statement and answer the questions that follow.
HTTP requests can be processed independently, while a web application can still maintain a user's shopping basket across several requests.

a) Describe how a session identifier in a cookie can link separate requests to a basket stored on the server. (3)

b) Explain what may happen to access to the existing basket if that cookie is deleted. In your answer, distinguish the identifier from the basket data. (2)

Question 8 Checking cached content

Suppose a browser holds a saved copy of a page and sends a conditional HTTP request containing an If-None-Match header. The server finds that the page has not changed.

a) Describe the purpose of the validator supplied in the request and identify the response the server should return. (3)

b) Explain how the browser uses this response and why it can reduce the amount of data transferred. (2)

Question 9 Stop and wait communication

A reliable data transfer protocol sends one data packet and waits for its acknowledgement before sending another. The link is fast, but the round-trip delay is long.

a) Explain why this arrangement can leave the link idle even when the sender has more data available. (3)

b) Describe how pipelining addresses this limitation without requiring a faster physical link. (2)

Question 10 TCP duplicate data

Suppose a TCP receiver obtains data that it has already accepted earlier in the same connection. The repeated data is not corrupted.

a) Explain how the receiver can recognise the repeated data and why it should not deliver it to the application again. (3)

b) State how the receiver should acknowledge this event and explain the purpose of that response. (2)

Question 11 Explicit congestion notification

Read the following statement and answer the questions that follow.
A router can warn an ECN-capable TCP connection about congestion by marking a packet before its queue overflows.

a) Describe how the congestion indication reaches the TCP sender and the response expected from the sender. (3)

b) Explain why such a warning can be useful even when the marked packet is delivered successfully. (2)

Question 12 Head of line blocking

Suppose packets are waiting in a single queue at a router input port. The first packet requires a busy output port, while a packet behind it requires an available output port.

a) Explain why the packet behind the first one may still be delayed. Name this condition. (3)

b) Describe one queue organisation that could reduce this problem. (2)

Question 13 Match action forwarding

An SDN switch has a rule that matches packets arriving from a guest network and destined for an internal payroll server. The specified action is to drop those packets.

a) Identify the match information and the action in this example. (2)

b) Explain how this decision uses more information than destination-only forwarding, and describe one additional action a match-action rule could specify. (3)

Question 14 The best effort service model

Read the following statement and answer the questions that follow.
A sender successfully hands a datagram to an IP network, but the receiving application does not necessarily obtain that datagram.

a) Explain why accepting the datagram does not create a guarantee of delivery or arrival order under a best-effort service model. (3)

b) Describe one mechanism an application or transport protocol could add to detect missing data and recover from it. (2)

Question 15 Recovery after a failed link

Suppose a routed network has an alternative path between two sites. When the active link fails, some traffic is temporarily lost before communication resumes.

a) Explain why an alternative physical path does not necessarily provide immediate recovery. (2)

b) Describe what must happen before routers consistently use the alternative path. In your answer, include failure detection and the updating of forwarding information. (3)

Question 16 Managing a withdrawn route

An organisation learns a route to an external network through a neighbouring BGP router. That neighbour later withdraws the route, and no alternative route is available.

a) Explain what the withdrawal tells the organisation and what should happen to the affected routing information. (3)

b) Explain why continuing to advertise the withdrawn route to other neighbours would create a network problem. (2)

Question 17 Polling and event notifications

Suppose a network manager checks router interface status at regular intervals. A brief interface failure occurs and is resolved between two checks.

a) Explain why periodic polling may fail to reveal the incident and how an event notification could help. (3)

b) State one limitation of relying only on event notifications and explain why combining the two methods is useful. (2)

Question 18 Limits of error detection

Read the following statement and answer the questions that follow.
A frame passes an error-detection check, but this does not prove with absolute certainty that every transmitted bit arrived unchanged.

a) Explain why an error-detection mechanism may fail to detect some error patterns. (2)

b) Distinguish detecting an error from correcting it, and describe one way communication can recover after a damaged frame is detected. (3)

Question 19 Access to a shared medium

Suppose many devices share one communication channel. One design allows devices to contend for access, while another gives devices permission to transmit in turn.

a) Explain what happens when transmissions overlap in the contention-based design and how devices can try again. (3)

b) Explain why a turn-taking design may perform more predictably under heavy load. In your answer, mention one overhead introduced by turn-taking. (2)

Question 20 Traffic within a data centre

A hosted application uses a web server, an application server and a database server. A single request from an Internet user causes several exchanges between these servers.

a) Explain why a data centre must support substantial traffic between its servers as well as traffic to and from Internet users. (3)

b) Describe how providing multiple internal paths can support this traffic, and state one benefit when an internal link fails. (2)